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Completed	Tuesday, 20 October 2026, 1:24 PM
Duration	9 mins 13 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

- a. The search is considered unsuccessful ✓
- b. The last element is returned
- c. The first element is returned
- d. An error is raised

Question **2**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

- a. By using a hash function
- b. By comparing each element sequentially
- c. By starting from the first element
- d. By dividing the list length by two ✓

Question **3**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

- a. O(log n)
- b. O👎 ✓
- c. O(n log n)
- d. O(1)

Question **4**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

a. n/2
b. log n
c. n ✓
d. n-1

Question **5**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

a. 90 and 99 ✓
b. 89 and 94
c. 90 and 100
d. 94 and 99

Question **6**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

a. When the list is sorted ✓
b. When the list is unsorted
c. When the list is very small
d. When the list contains duplicate elements

Question **7**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

- a. Linear search ✓
- b. Binary search
- c. Both (1) & (3)
- d. Hash search

Question **8**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

- a. O(1) ✓
- b. O(n log n)
- c. O🤜
- d. O(log n)

Question **9**

Correct

Mark 1.00 out of 1.00

What is the first step in binary search?

- a. Compare the target element with the first element in the list
- b. Compare the target element with the middle element in the list ✓
- c. Divide the list into two equal parts
- d. Sort the list

Question **10**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

a. The search continues in the right sublist
b. The search continues in the left sublist
c. The list is sorted
d. The search ends successfully ✓

Question **11**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. Binary search algorithm is not efficient when the data elements more than 1500 ✓
b. There must be a mechanism to access middle element directly
c. Must use a sorted array
d. Requirement of sorted array is expensive when a lot of insertion and deletions are needed

Question **12**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

a. Binary search does not require dividing the list
b. Binary search can find multiple instances of the target element
c. Binary search has a lower time complexity on large, sorted lists ✓
d. Binary search works on unsorted lists

Question **13**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search is more efficient than binary search on large lists.
b. Linear search divides the list into halves.
c. Linear search requires the list to be sorted.
d. Linear search can be applied to both sorted and unsorted lists. ✓

Question **14**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

- a. The search continues from the beginning
- b. The search stops
- c. The list is sorted
- d. The search continues in the left or right sublist ✓

Question **15**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

- a. It works on unsorted lists
- b. It can be applied only if the list is sorted ✓
- c. It compares elements sequentially
- d. It always starts from the beginning of the list